Unit Test - Form A

Introduction to Matter

STUDENT ASSESSMENT

Name: ____________________ Date: __________ Period: _____

Directions: Select the best answer for each question. Mark one choice only. Complete this test after all lessons and the cumulative review.

1. Which sample is matter?

A. Visible light

B. Sound from a speaker

C. Air in a sealed syringe

D. A reflected image

2. A particle model contains two particle types evenly distributed. What is the best classification?

A. Element

B. Compound

C. Heterogeneous mixture

D. Homogeneous mixture

3. Which statement describes a limitation of a typical particle model?

A. Particle size and spacing are not shown to scale.

B. It can never show more than one particle type.

C. It proves every property of the sample.

D. Its symbols are literal photographs.

4. Which evidence best supports that a clear solution is a mixture?

A. It is colorless.

B. It occupies a beaker.

C. It has a measurable mass.

D. A physical process recovers two components.

5. How should a clean copper wire be classified?

A. Element

B. Compound

C. Homogeneous mixture

D. Heterogeneous mixture

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6. How should pure water be classified?

A. Element

B. Compound

C. Homogeneous mixture

D. Heterogeneous mixture

7. Why is well-mixed air not a pure substance?

A. It has no mass.

B. It is invisible.

C. It contains multiple substances.

D. It cannot fill a container.

8. Granite with visibly different mineral regions is a

A. compound

B. element

C. homogeneous mixture

D. heterogeneous mixture

9. Which property is physical and intensive?

A. Mass

B. Density

C. Volume

D. Flammability

10. Which property is chemical?

A. Color

B. Melting point

C. Conductivity

D. Flammability

A

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11. Which property depends on sample amount?

A. Mass

B. Density

C. Melting point

D. Boiling point

12. Which process is a physical change?

A. Paper burns.

B. Iron rusts.

C. Ice melts.

D. A compound decomposes.

13. Which observation is decisive evidence of a chemical change?

A. The sample changes shape.

B. The sample becomes smaller.

C. A liquid boils.

D. New substances with different compositions form.

14. Why do bubbles not always prove a chemical reaction?

A. Boiling can also produce bubbles.

B. Gases are not matter.

C. Bubbles contain no particles.

D. All bubbles are physical objects.

15. Which method separates insoluble sand from water?

A. Distillation

B. Filtration

C. Chromatography

D. Magnetism

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16. Which method should be used to collect both water and salt from saltwater?

A. Magnetic separation

B. Simple filtration

C. Decantation only

D. Distillation

17. Which method tests whether marker ink contains multiple dyes?

A. Chromatography

B. Filtration

C. Magnetism

D. Decantation

18. What is the best first step for dry iron filings mixed with salt?

A. Boil the mixture.

B. Add acid.

C. Use a magnet.

D. Filter the dry mixture.

19. A 120.0 g sample yields 92.0 g liquid and 27.6 g solid. What is percent recovery?

A. 76.7%

B. 99.7%

C. 100.4%

D. 119.6%

20. Why may solid mass decrease when heating occurs in an open container?

A. Matter is destroyed.

B. The balance creates gas.

C. A gaseous product may leave the container.

D. Conservation applies only to solids.